

VISTA

Retail shelf inspection and inventory visibility

Supporting brand presentation and sales

Native iPhone vision and on-device AI models.
Server-side agentic AI on Google Cloud.
Historical data analysis and continuous improvement.

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September 2026



Concept illustration

Teams need to understand what is happening across stores

To better manage inventory, brand presentation and sales.

Before the visit



After the visit



**What is present?
How many facings?
What changed?**

Efficient capture and analysis,
automated as much as possible.

The technical challenge starts with
complete, readable shelf images.

Original field examples supplied with the project

Guided shelf scanning

Define the shelf once. Capture repeatable evidence with less inspector effort.



Pharmacy capture workflow, concept illustration.

01 Define the shelf

Save a reference image, target area and versioned shelf profile.

02 Guide the capture

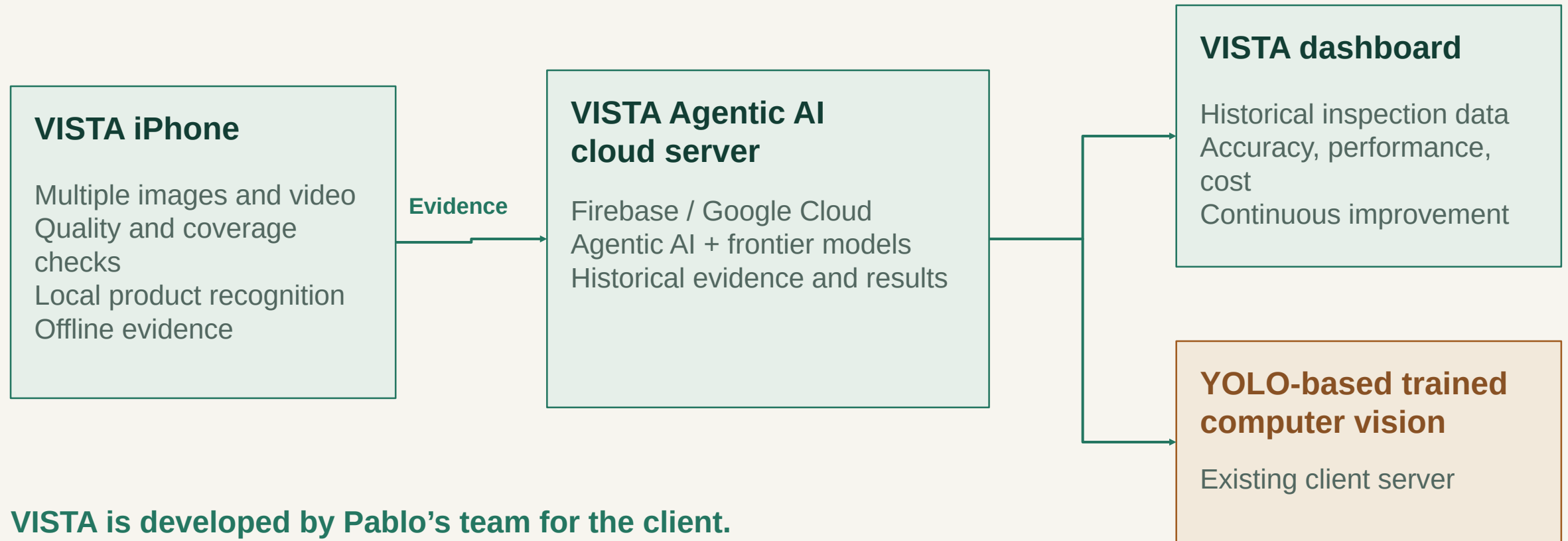
Capture multiple images and video.
Check sharpness, stability and coverage.

03 Analyze and review

Connect local findings, server analysis and human review to the same evidence.

Solution architecture

VISTA integrates guided capture and agentic AI with the client's existing YOLO-based server.



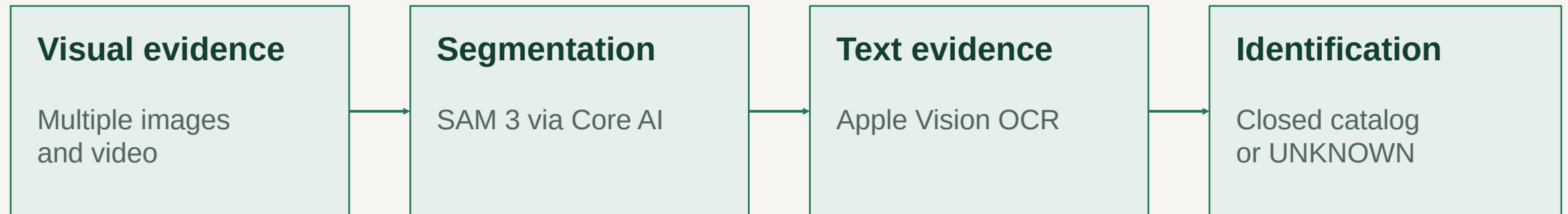
VISTA is developed by Pablo's team for the client.
The client's existing YOLO-based server is a separate system.

On-device computer vision

Local processing supports capture quality and a constrained first interpretation.

Live guidance

AVFoundation, Apple Vision and Core Motion check stability, sharpness, location and coverage.

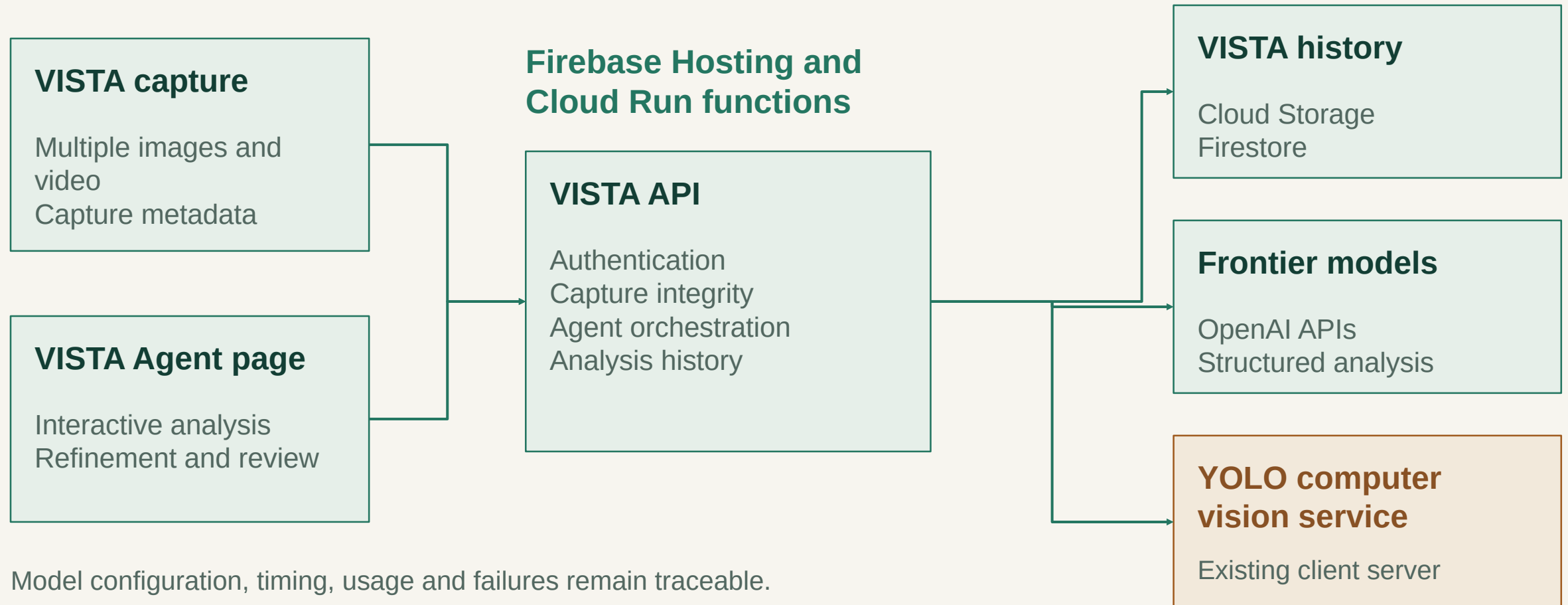


Recognition starts after capture finalization.
Coverage-aware aggregation suppresses repeated sightings.

SwiftUI
SQLite + SHA-256 audit

VISTA Agentic AI cloud server

VISTA on Google Cloud connects frontier models, the client's YOLO-based server and history.



Model configuration, timing, usage and failures remain traceable.

Historical evaluation and continuous improvement

Track facing-count accuracy, speed, total cost and model versions over time.

VISTA

CONCEPT VIEW

Inspection comparison

Source evidence



 1 image attached

 1 video attached

iPhone Vision

Products

—

Counts

—

Uncertainty

—

YOLO

Products

—

Counts

—

Uncertainty

—

Agentic AI

GPT or other models

Products

—

Counts

—

Reasoning

—

Human

Verified labels

—

Corrections

—

Review status

—

Historical evaluation

Metric	Earlier runs	Recent runs	Latest run
Accuracy vs human review	—	—	—
Processing time	—	—	—
Cost per inspection	—	—	—

Compare the same evidence across approaches and track changes over time

YOLO findings come from the client’s existing server. VISTA compares all four approaches.

Roadmap: compare, tune and re-evaluate

Use historical inspections to improve facing counts, speed and total cost.

iPhone Vision	Client YOLO	Agentic AI	Human review
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Accuracy	Shelf facings by product; count error vs independent human review.
Speed	End-to-end seconds per inspection, including human review time.
Total cost	Per inspection and total run; include training and labeling effort.
Models used	Exact model ID/version and configuration; human reviewer/protocol.

1 Establish baseline

Same images/video;
independent human counts.

2 Tune the best options

Models, prompts, thresholds
and routing between layers.

3 Re-evaluate over time

Replay held-out cases;
compare each new version.

Solution architecture in practice

Pablo Elustondo / Architecture, hands-on engineering and field evaluation

Business and workflow

Reframed the problem around controllable capture and separated onboarding from recurring inspection.

Architecture and delivery

Connected native iPhone evidence with cloud services, explicit component responsibilities and reviewable contracts.

Hands-on AI engineering

Built and evolved the VISTA Agentic AI cloud server with model integration, refinement and run diagnostics.

Validation and improvement

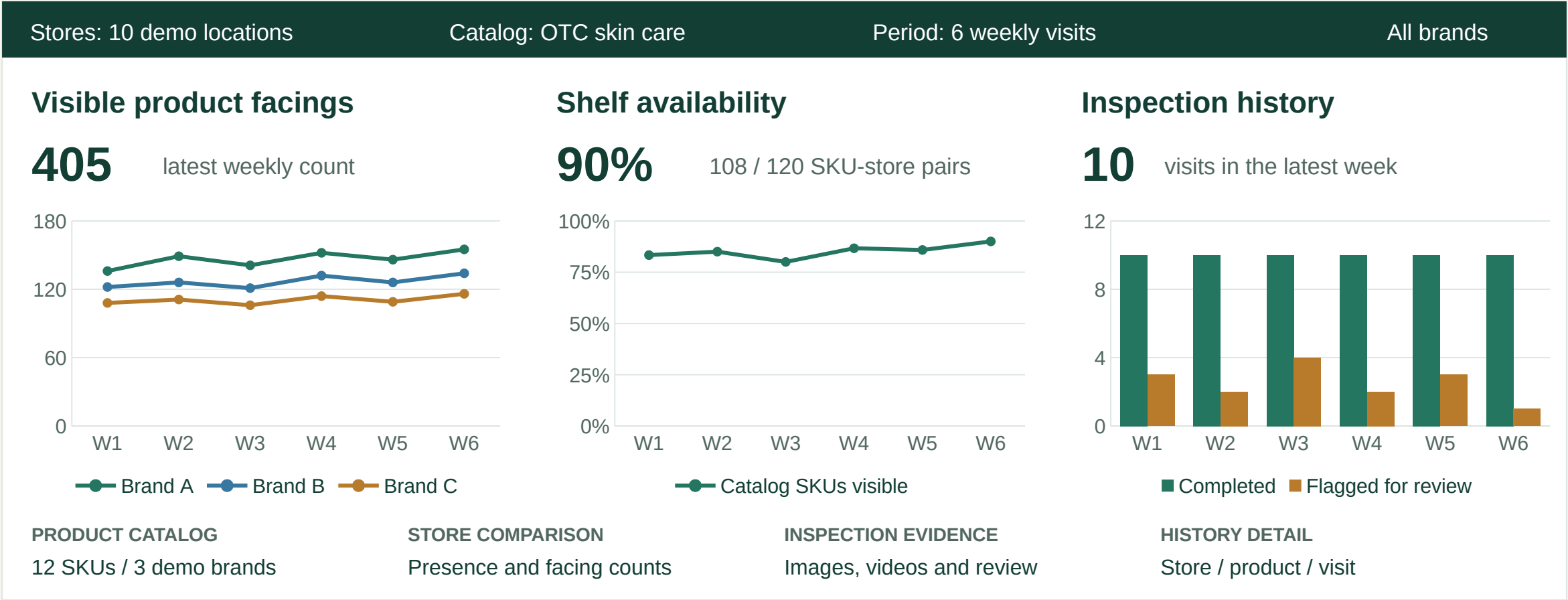
Ran real shelf experiments and shaped benchmarks, acceptance criteria and the next implementation increments.

Collaborative delivery: native iOS development with Williams Martinez.

VISTA Mission Control

Near-term roadmap. Not yet in development.

Illustrative dashboard data



Trends across visits, with drill-down to stores, catalog products and inspection evidence.